A FUNCTOR DEFINED ON FINITELY PRESENTED MODULES EXTENDS UNIQUELY TO A FUNCTOR COMMUTING WITH FILTERED COLIMITS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 0FWY: sources/stacks-chow/tags.txt, source lines 19863–19863. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0G8P: sources/stacks-chow/tags.txt, source lines 20274–20274. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00HA: sources/stacks-chow/tags.txt, source lines 636–636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a ring. Let $\mathcal{B}$ be a category having filtered colimits. Let $F : Mod_A^{fp} \to \mathcal{B}$ be a functor. Then F extends uniquely to a functor $F' : Mod_A \to \mathcal{B}$ which commutes with filtered colimits.*

Proof. This follows from Categories, Lemma 0FWY. To see that the lemma applies observe that finitely presented A-modules are categorically compact objects of Mod_A by Algebra, Lemma 0G8P. Also, every A-module is a filtered colimit of finitely presented A-modules by Algebra, Lemma 00HA. □

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